KAIZEN MATHS

Lesson starter

Permutations and Combinations

Counting Foundations

Name _____

S1

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

Evaluate $\frac{6!}{4!}$.

S2

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

A student can choose from 5 shirts, 2 pairs of trousers, and 2 pairs of shoes. How many different outfits are possible?

S3

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

How many ways can 5 different students stand in a line?

S4

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

From 11 students, how many different groups of 2 can be chosen?

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STA-70FE-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Permutations and Combinations

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Name _____

EA1

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

Evaluate $\frac{7!}{3!}$.

EA2

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A student can choose from 6 shirts, 4 pairs of trousers, and 4 pairs of shoes. How many different outfits are possible?

Show your working

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Exit ticket

Permutations and Combinations

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Name _____

EB1

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

How many ways can 7 different students stand in a line?

Show your working

KAIZEN MATHS

Exit ticket

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Evaluate $\frac{7!}{3!}$.

EA2

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A student can choose from 6 shirts, 4 pairs of trousers, and 4 pairs of shoes. How many different outfits are possible?

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Show your working

KAIZEN MATHS

Exit ticket

Permutations and Combinations

Counting Foundations

Name _____

EB1

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

How many ways can 7 different students stand in a line?

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Show your working

Teacher answers & working

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S1

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

Evaluate $\frac{6!}{4!}$.

Answer:

30

Working:

Use factorial notation: 6! means 6 × 5 × 4 × … × 1.

$\frac{6!}{4!} = \frac{6 \times 5 \times 4!}{4!}$

Cancel the common 4! factor.

$6 \times 5 = 30$

S2

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

A student can choose from 5 shirts, 2 pairs of trousers, and 2 pairs of shoes. How many different outfits are possible?

Answer:

20

Working:

There are three independent choices, so use the product rule.

$5 \times 2 \times 2 = 20$

Each shirt can go with each pair of trousers and each pair of shoes.

S3

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

How many ways can 5 different students stand in a line?

Answer:

120

Working:

The first position can be filled in n ways, then one fewer choice remains each time.

$5 \times 4 \times 3 \times \ldots \times 1 = 5!$

$5! = 120$

Teacher answers & working

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S4

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

From 11 students, how many different groups of 2 can be chosen?

Answer:

55

Working:

This is a selection, so order does not matter.

$^{11}C_{2} = \frac{11!}{2!9!}$

$^{11}C_{2} = \frac{39,916,800}{2 \times 362,880} = 55$

EA1

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

Evaluate $\frac{7!}{3!}$.

Answer:

840

Working:

Use factorial notation: 7! means 7 × 6 × 5 × … × 1.

$\frac{7!}{3!} = \frac{7 \times 6 \times 5 \times 4 \times 3!}{3!}$

Cancel the common 3! factor.

$7 \times 6 \times 5 \times 4 = 840$

EA2

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

A student can choose from 6 shirts, 4 pairs of trousers, and 4 pairs of shoes. How many different outfits are possible?

Answer:

96

Working:

There are three independent choices, so use the product rule.

$6 \times 4 \times 4 = 96$

Each shirt can go with each pair of trousers and each pair of shoes.

Teacher answers & working

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EB1

Decide whether the question is asking for an arrangement or a selection, then calculate the number of possibilities.

How many ways can 7 different students stand in a line?

Answer:

5,040

Working:

The first position can be filled in n ways, then one fewer choice remains each time.

$7 \times 6 \times 5 \times \ldots \times 1 = 7!$

$7! = 5{,}040$